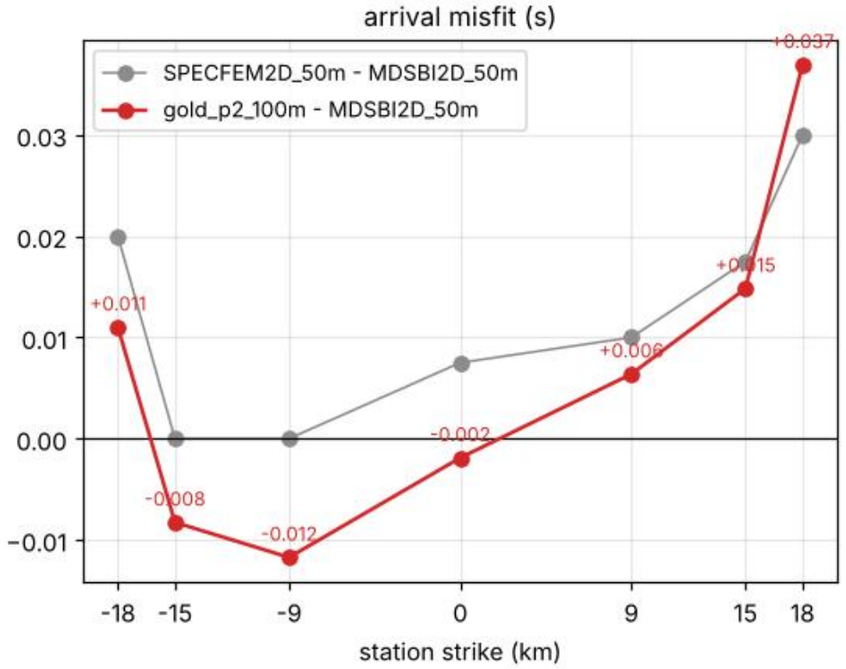
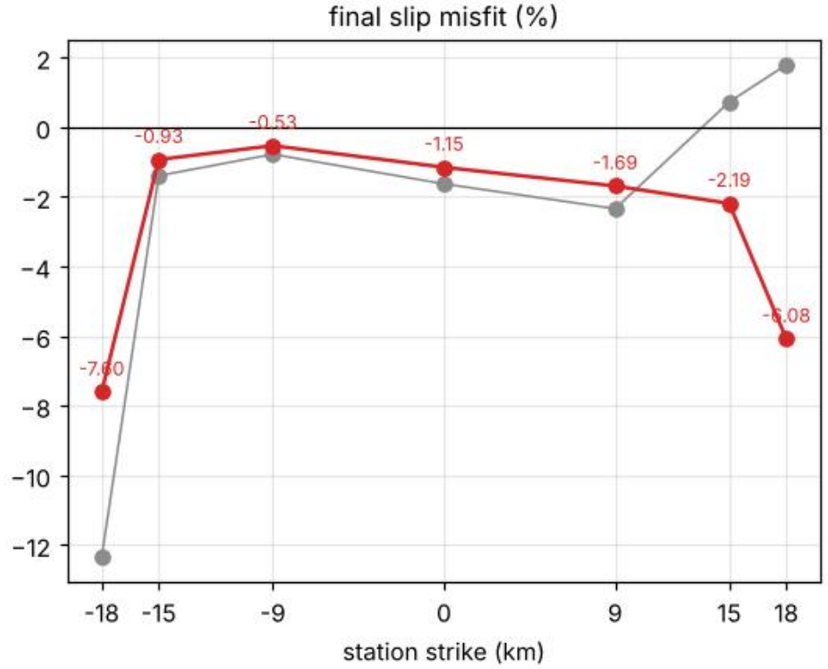
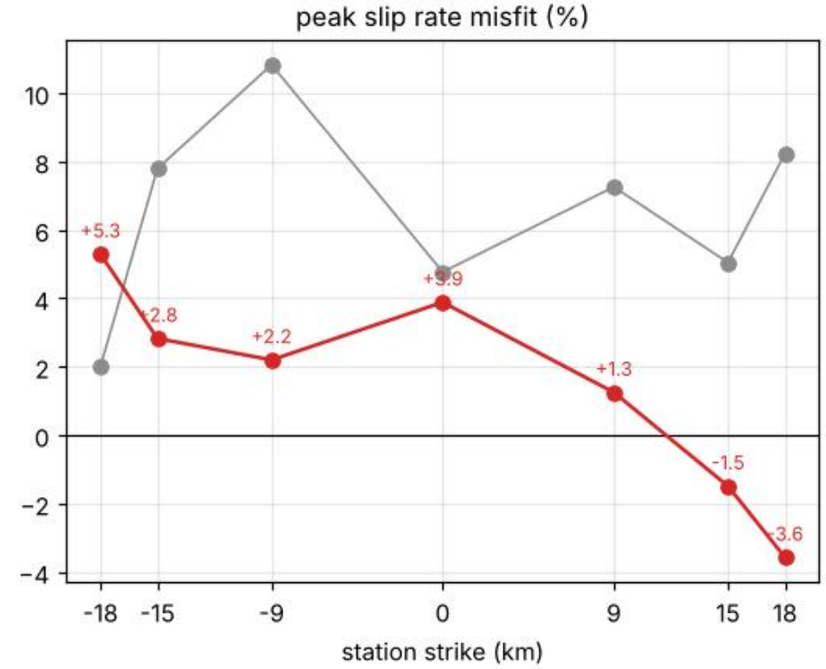
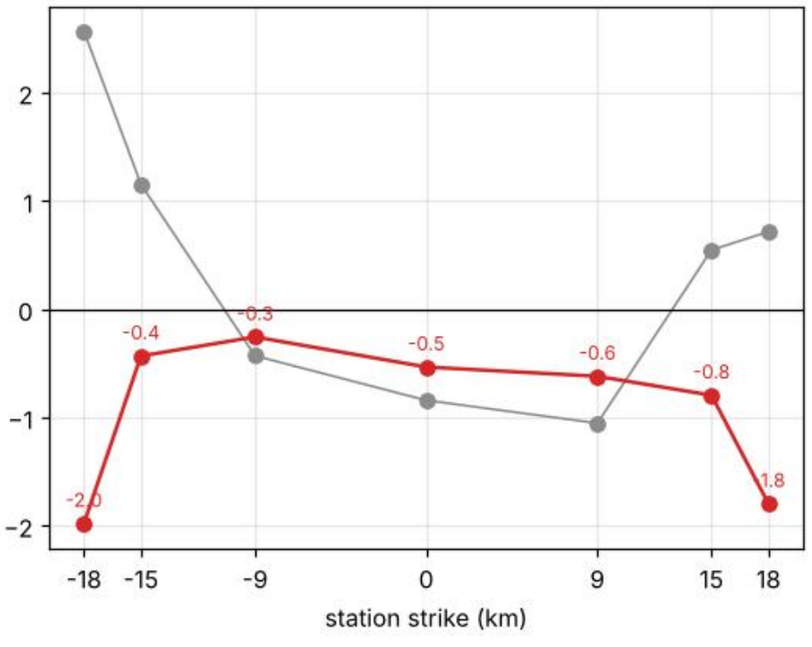
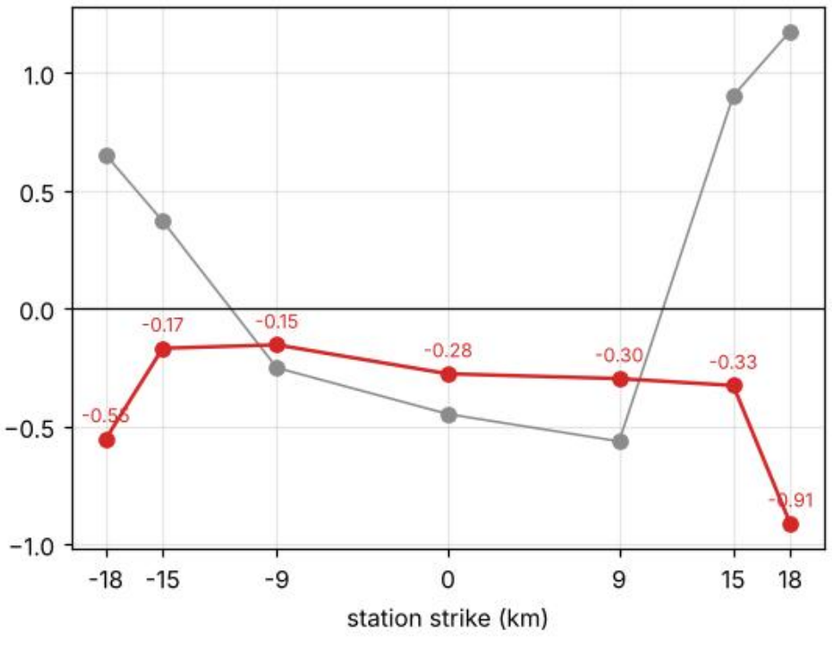
SCEC TPV105-2D: runs and references minus MDSBI2D_50m at the 7 on-fault stations (labels: gold_p2_100m)











window 0 to 12.5 s (the shortest record)
misfits against MDSBI2D_50m

gold p2 100 m, commit b0ce3cb6, cloud 4 ranks
grey = SPECFEM2D 50 m minus MDSBI 50 m

SCEC TPV105-2D station x2_-18_x3_7.5 (strike -18 km)
gold_p2_100m vs MDSBI2D_50m: arrival +0.011 s, final slip -7.60 %, peak slip rate +5.3 %, peak T -2.0 %, final p -0.55 MPa

(a) slip rate (m/s)

(b) slip rate (m/s), MDSBI2D_50m arrival 3.620 s -0.5 / +2.5 s

(c) slip (m)

(d) shear stress (MPa)

(e) effective normal stress, compression positive (MPa)

(f) shear / effective normal stress

(g) temperature (K)

(h) pore pressure (MPa)

(i) state variable psi

time (s)

time (s)

time (s)

MDSBI2D_50m SPECFEM2D_50m gold_p2_100m

SCEC TPV105-2D station x2_-15_x3_7.5 (strike -15 km)
gold_p2_100m vs MDSBI2D_50m: arrival -0.008 s, final slip -0.93 %, peak slip rate +2.8 %, peak T -0.4 %, final p -0.17 MPa

(a) slip rate (m/s)

(b) slip rate (m/s), MDSBI2D_50m arrival 2.150 s -0.5 / +2.5 s

(c) slip (m)

(d) shear stress (MPa)

(e) effective normal stress, compression positive (MPa)

(f) shear / effective normal stress

(g) temperature (K)

(h) pore pressure (MPa)

(i) state variable psi

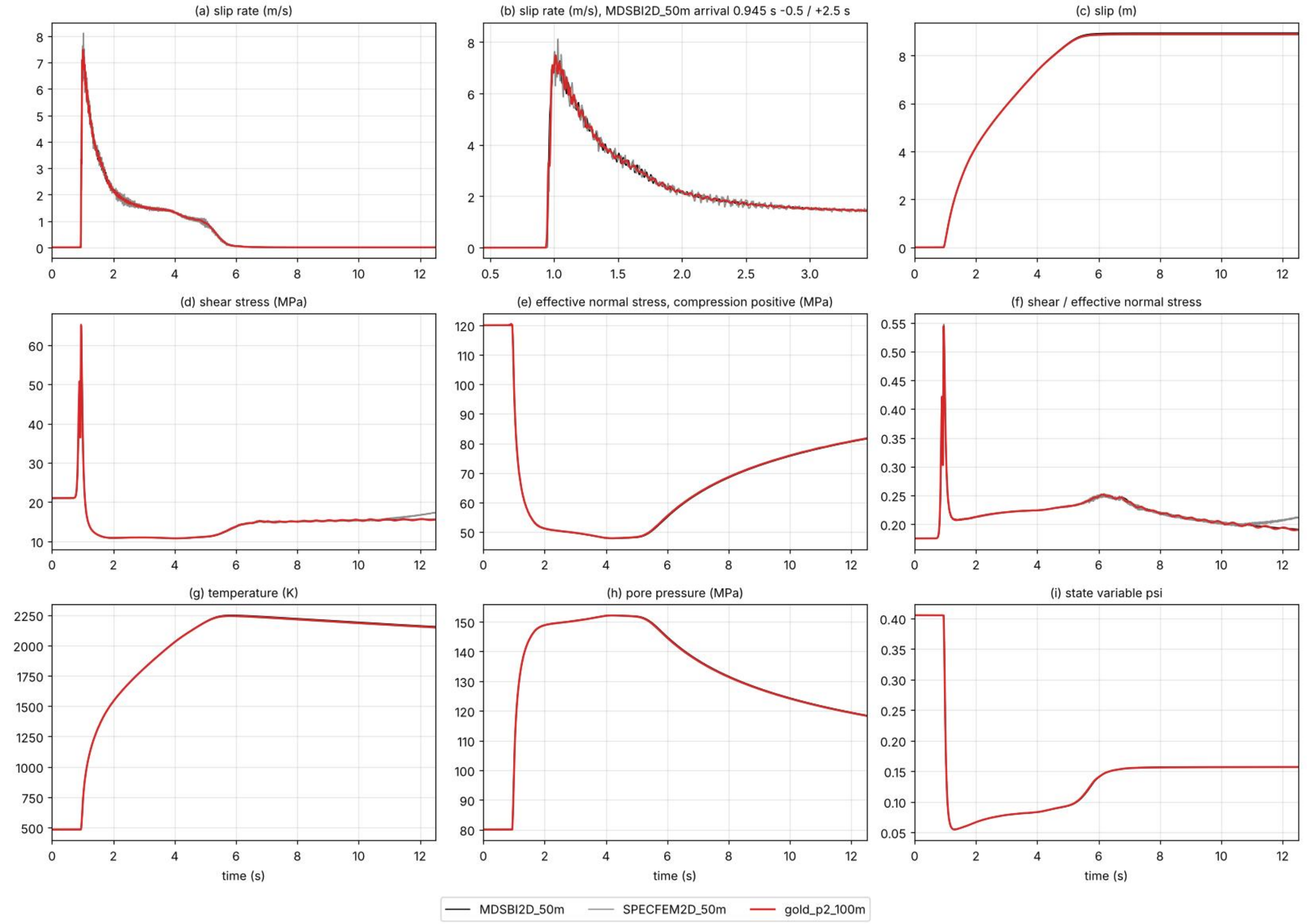
time (s)

time (s)

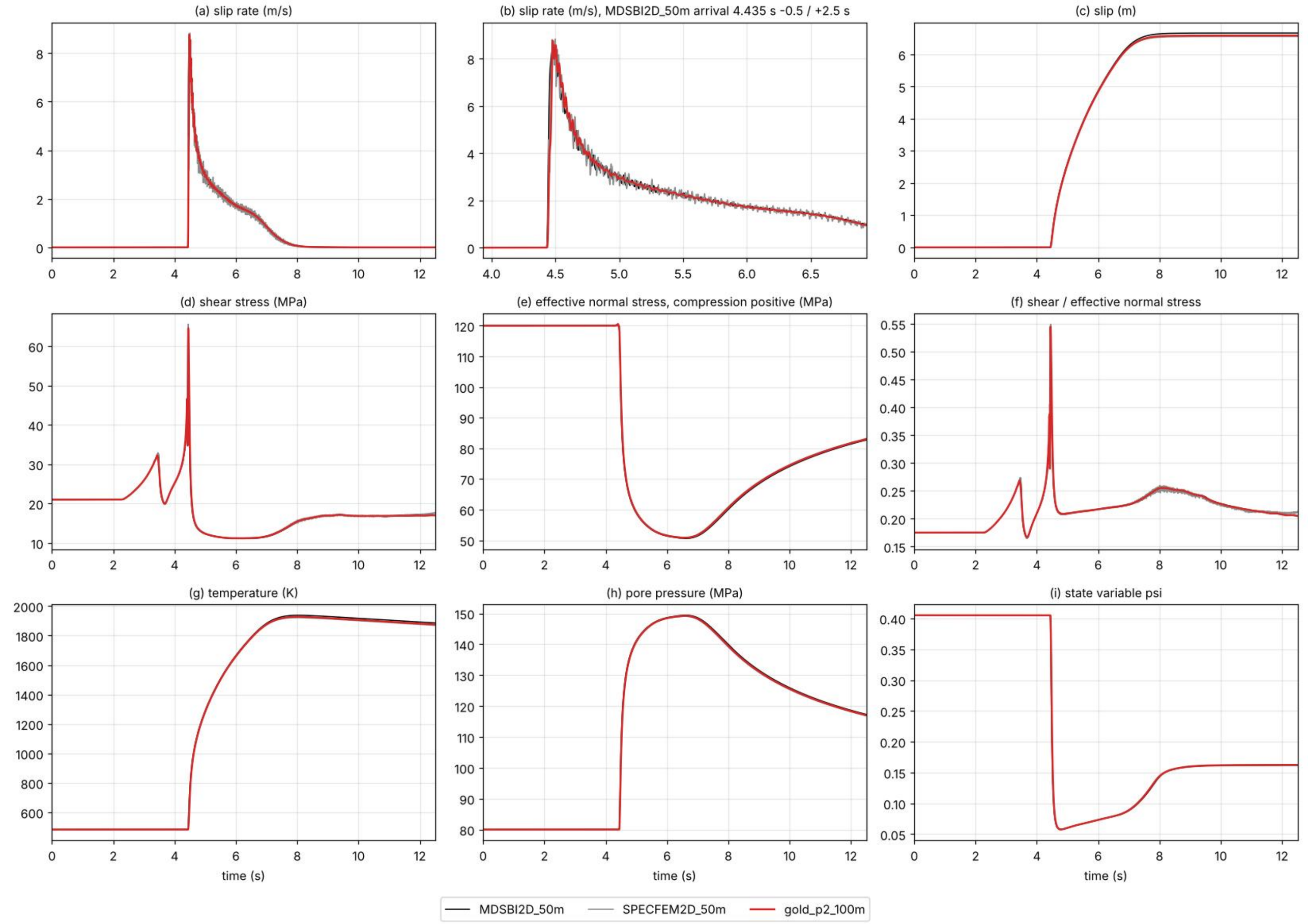
time (s)

MDSBI2D_50m SPECFEM2D_50m gold_p2_100m

SCEC TPV105-2D station x2_-9_x3_7.5 (strike -9 km)
gold_p2_100m vs MDSBI2D_50m: arrival -0.012 s, final slip -0.53 %, peak slip rate +2.2 %, peak T -0.3 %, final p -0.15 MPa

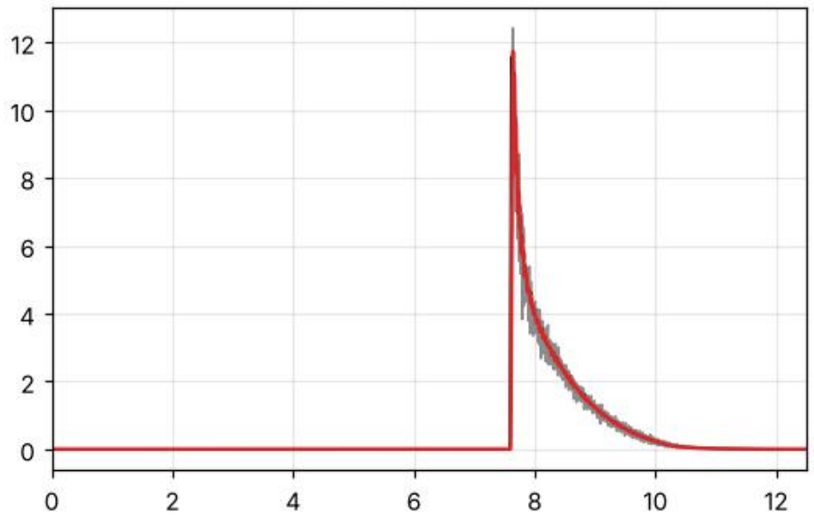


SCEC TPV105-2D station x2_0_x3_7.5 (strike 0 km)
gold_p2_100m vs MDSBI2D_50m: arrival -0.002 s, final slip -1.15 %, peak slip rate +3.9 %, peak T -0.5 %, final p -0.28 MPa

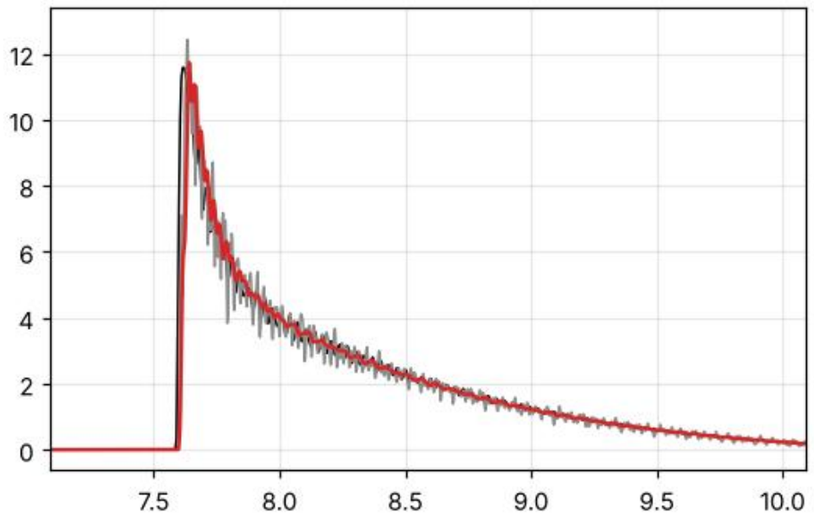


SCEC TPV105-2D station x2_9_x3_7.5 (strike 9 km)
gold_p2_100m vs MDSBI2D_50m: arrival +0.006 s, final slip -1.69 %, peak slip rate +1.3 %, peak T -0.6 %, final p -0.30 MPa

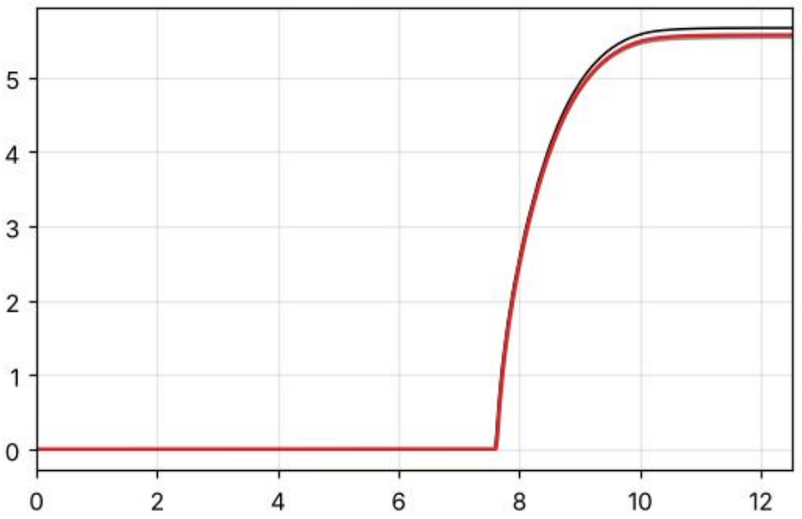
(a) slip rate (m/s)



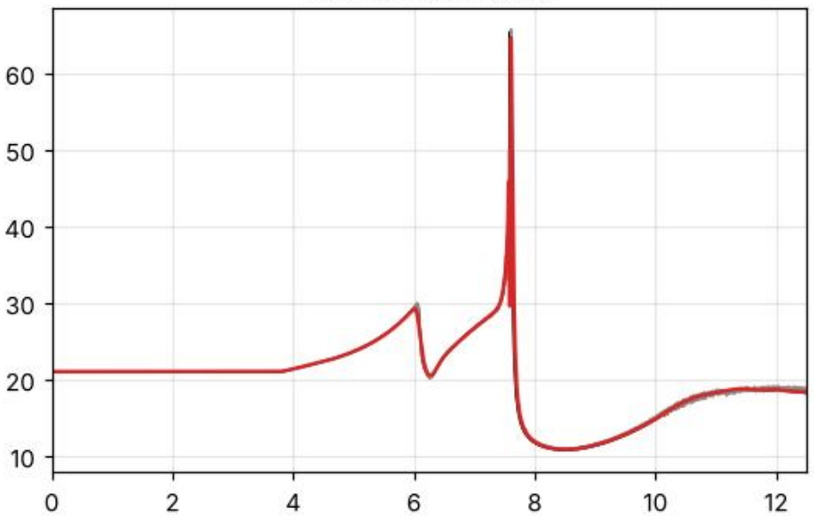
(b) slip rate (m/s), MDSBI2D_50m arrival 7.590 s -0.5 / +2.5 s



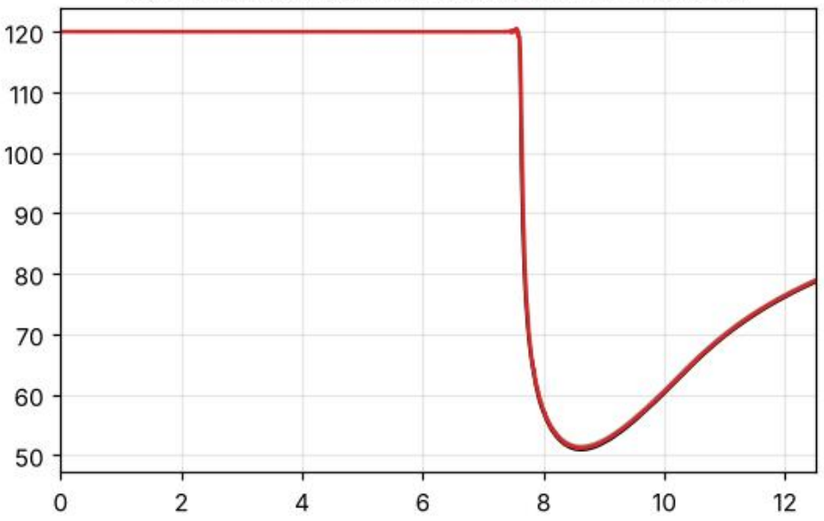
(c) slip (m)



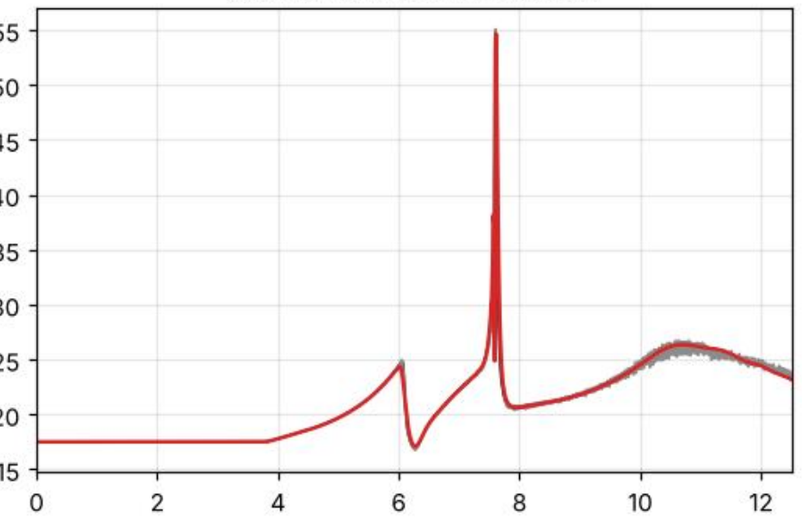
(d) shear stress (MPa)



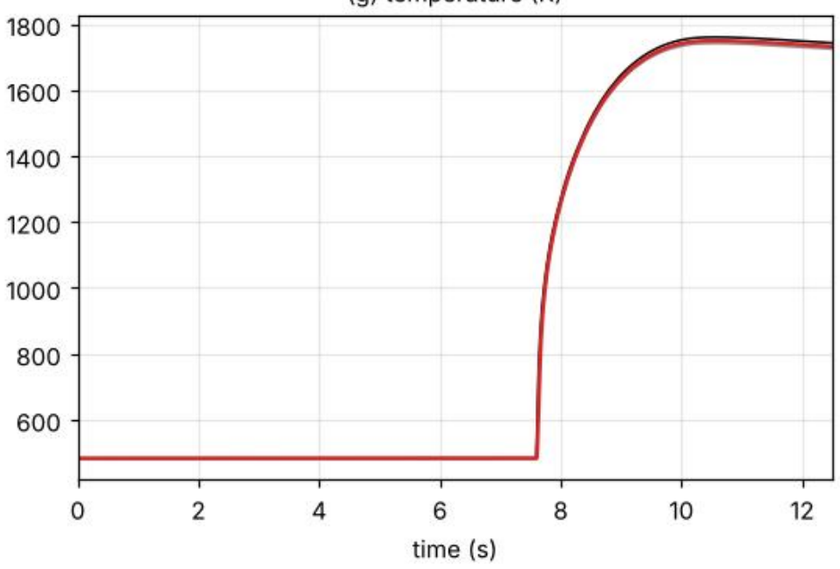
(e) effective normal stress, compression positive (MPa)



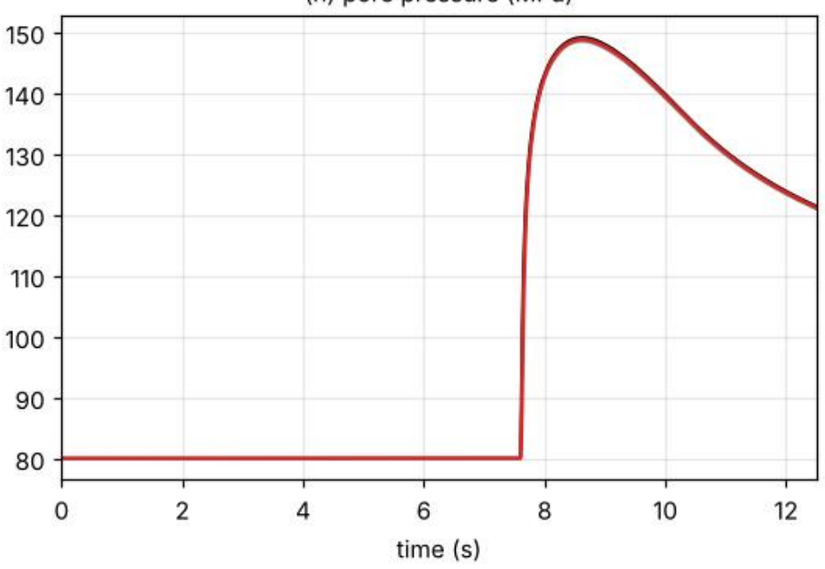
(f) shear / effective normal stress



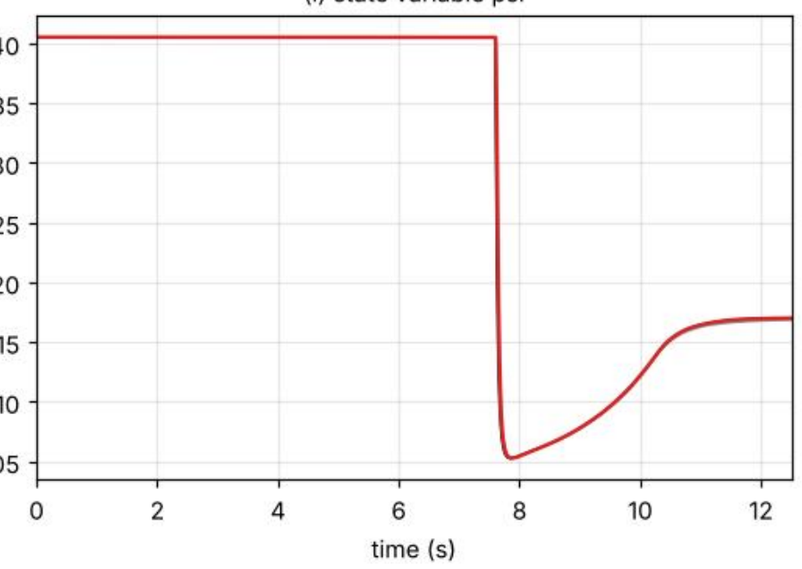
(g) temperature (K)



(h) pore pressure (MPa)



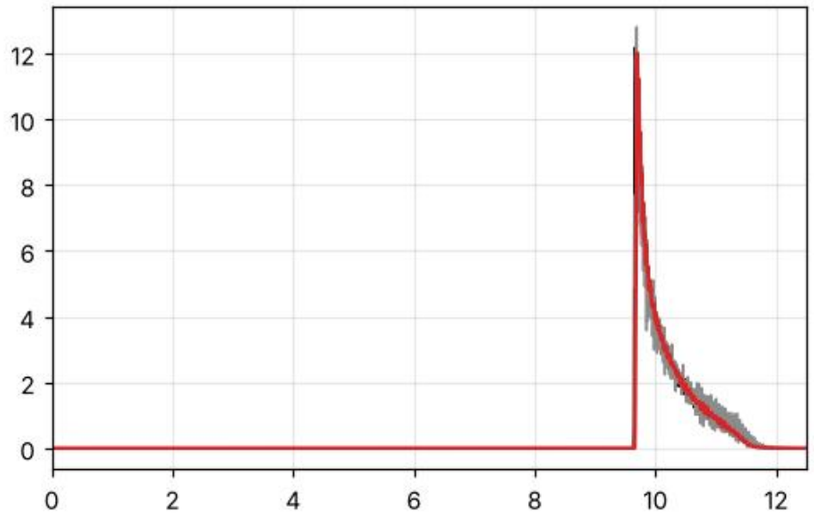
(i) state variable psi



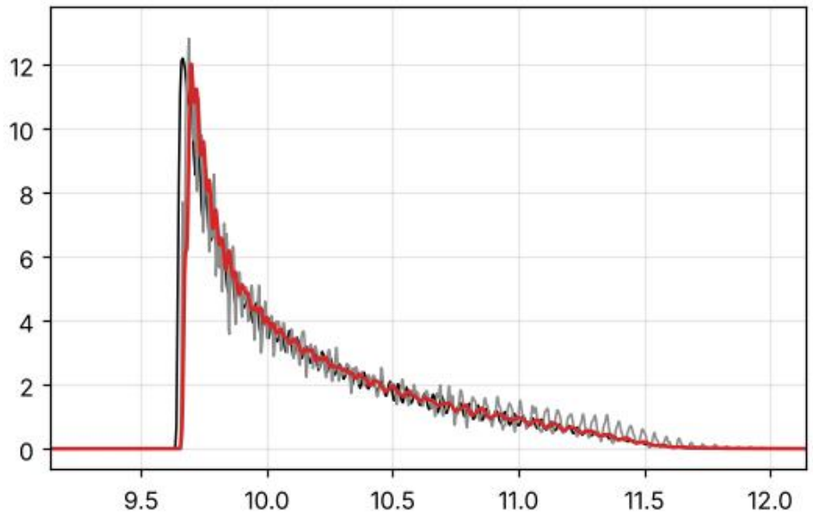
MDSBI2D_50m SPECFEM2D_50m gold_p2_100m

SCEC TPV105-2D station x2_15_x3_7.5 (strike 15 km)
gold_p2_100m vs MDSBI2D_50m: arrival +0.015 s, final slip -2.19 %, peak slip rate -1.5 %, peak T -0.8 %, final p -0.33 MPa

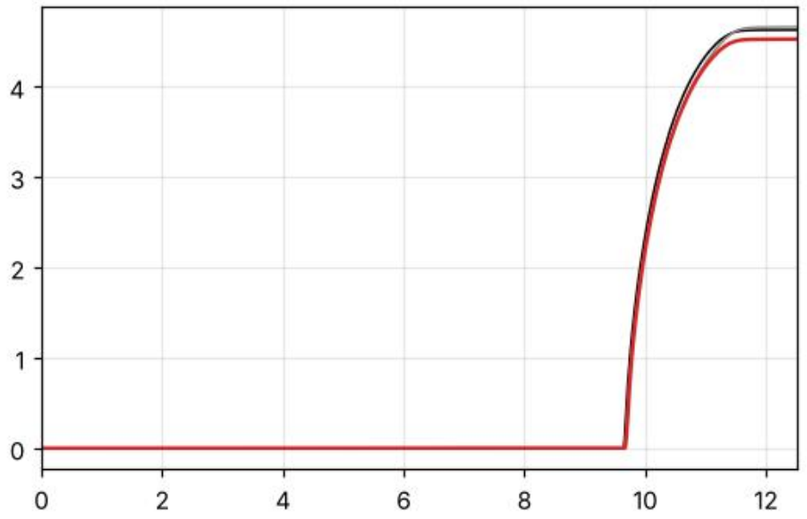
(a) slip rate (m/s)



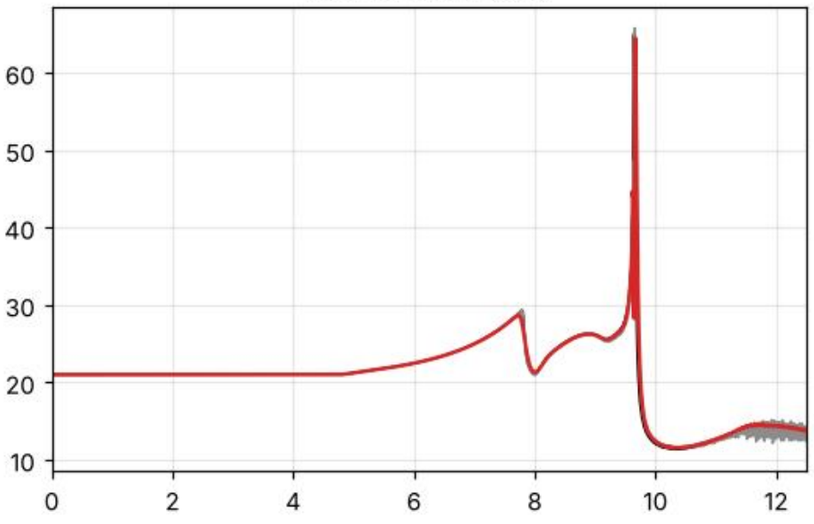
(b) slip rate (m/s), MDSBI2D_50m arrival 9.640 s -0.5 / +2.5 s



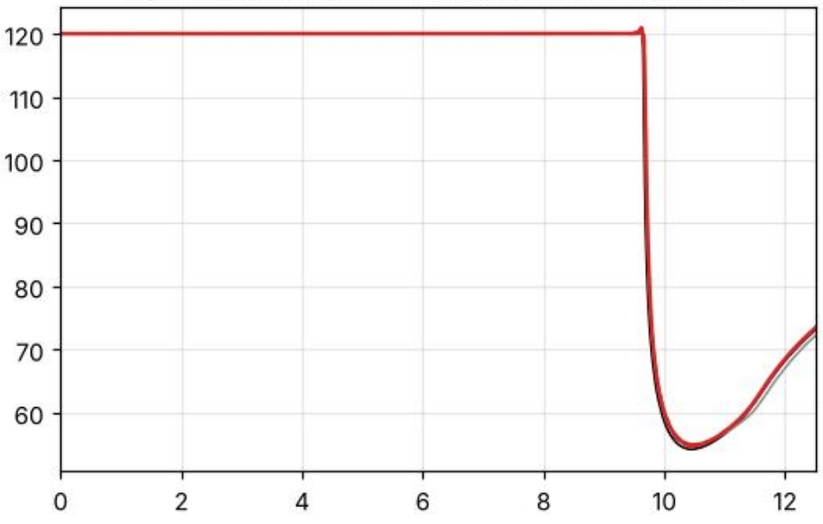
(c) slip (m)



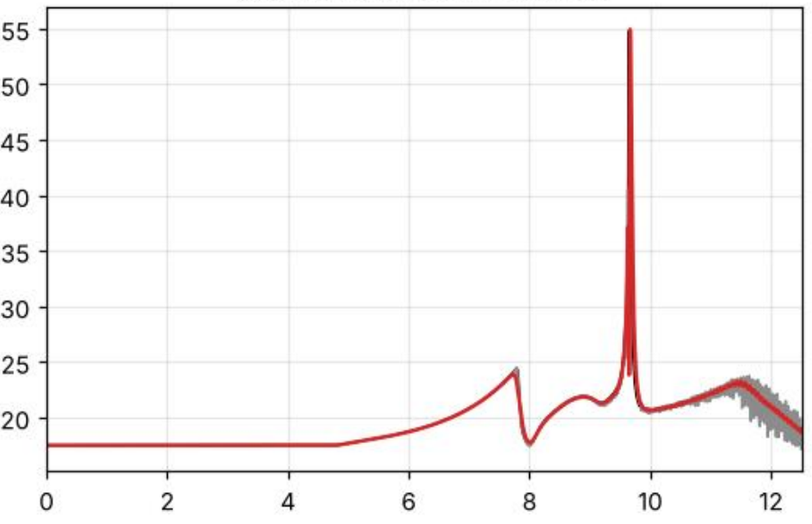
(d) shear stress (MPa)



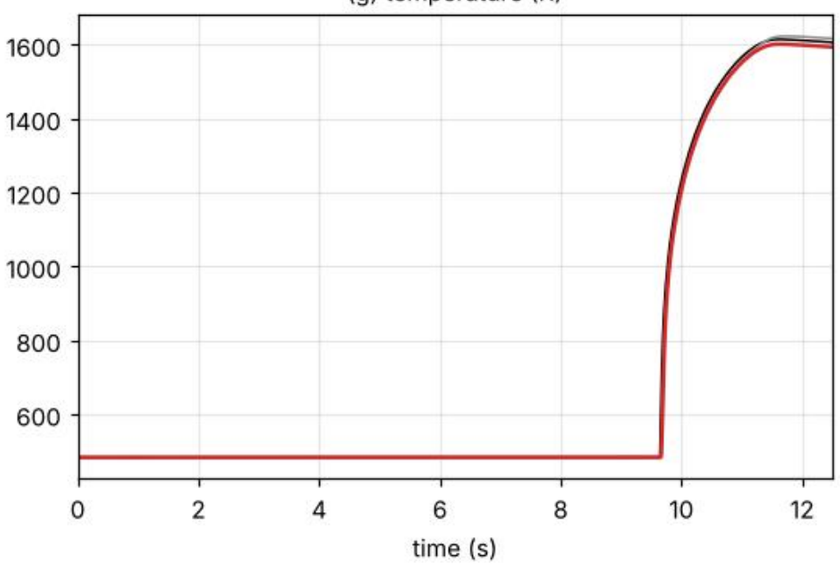
(e) effective normal stress, compression positive (MPa)



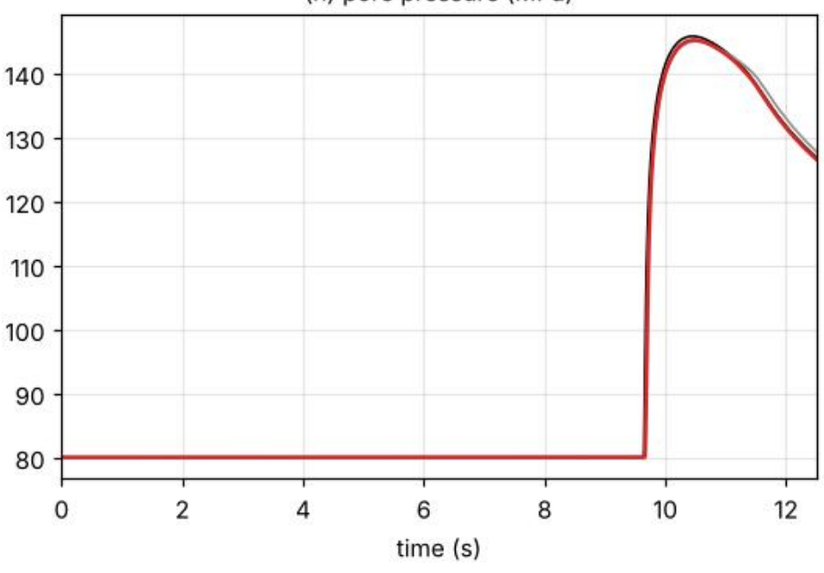
(f) shear / effective normal stress



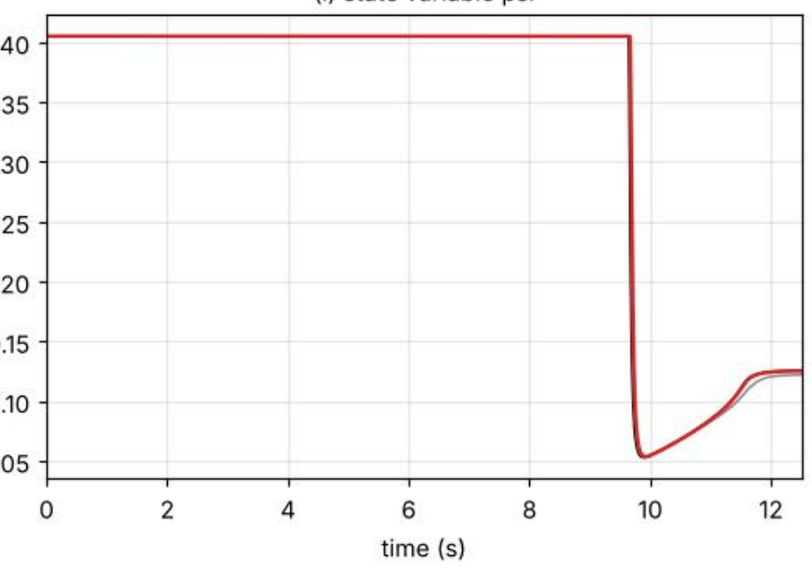
(g) temperature (K)



(h) pore pressure (MPa)



(i) state variable psi



MDSBI2D_50m SPECFEM2D_50m gold_p2_100m

SCEC TPV105-2D station x2_18_x3_7.5 (strike 18 km)
gold_p2_100m vs MDSBI2D_50m: arrival +0.037 s, final slip -6.08 %, peak slip rate -3.6 %, peak T -1.8 %, final p -0.91 MPa

(a) slip rate (m/s)

(b) slip rate (m/s), MDSBI2D_50m arrival 10.770 s -0.5 / +2.5 s

(c) slip (m)

(d) shear stress (MPa)

(e) effective normal stress, compression positive (MPa)

(f) shear / effective normal stress

(g) temperature (K)

(h) pore pressure (MPa)

(i) state variable psi

time (s)

time (s)

time (s)

MDSBI2D_50m SPECFEM2D_50m gold_p2_100m